

Fitting and choosing radiative-transfer emulators for the retrieval

Yitzchak Shmalo*

Abstract

Radiative-transfer emulators are fitted to a code’s outputs but used through an inverse, the retrieval of surface reflectance. We give the retrieval error exactly in terms of the component errors and show that the forward and retrieval rankings of two emulators differ by a change of measure with density proportional to t^2/q^4 . Training on the retrieval error is regression under a known, unbounded covariate shift whose price in effective sample size can be computed before any training. On the EMIT table, networks trained on the square root of the retrieval weights floored at a hundredth of the median transmission lower the radiance error by 8% and the 95th percentiles of the retrieval error by 32 to 44% on every one of ten splits. On the libRadtran benchmark of Mazid and Rishe the same weights lower the radiance error from 0.81% to 0.17% and the retrieval tail from 3.2 to 0.5 points on every seed, at a larger error in the spherical albedo. Two failed evaluations of the code in the EMIT table make every weighted network worse than the plain one, and refitting a convex stack on the retrieval error lowers its tail by a quarter.

1 Introduction

At each band the top-of-atmosphere radiance over a Lambertian surface of reflectance ρ is

$$L(\rho) = a + \frac{\rho t}{1 - \rho s},$$

with path radiance a , transmission t and spherical albedo s , and atmospheric correction recovers ρ from a measured radiance by inverting this relation [96]. Computing (a, t, s) with a radiative-transfer code such as MODTRAN [7], libRadtran [64, 33] or 6S [101] for every pixel is too slow, and the codes are replaced by emulators fitted to tables of their outputs [10, 81]. The emulators are fitted and compared by their forward error, the error in the emulated components or in the radiance they produce, but they are used through the inverse. An error in the emulated components enters the retrieved reflectance divided by the slope $\partial_\rho L = t/q^2$, $q = 1 - \rho s$, so that the same forward error costs little where the transmission is large and a great deal where it is small. On tables that reach into strong absorption the two criteria can rank emulators differently: on the EMIT atmospheric-correction table a convex stack of kernel and network emulators has the lowest radiance error of the emulators compared and a larger tail of the retrieval error than the best of its members [85].

The question here is what an emulator should be fitted to, and chosen by, when it is used through an inverse. The answer has three parts.

The retrieval error has an exact closed form in the three component errors (Lemma 1). To first order it is the forward error at ρ divided by the slope, and over a range of reflectances it is a Mahalanobis norm of the component errors with a metric that depends on the state only through

*Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel. yitzchak.shmalo@gmail.com.

(t, s) and is positive definite once three reflectances are used (Proposition 3). The forward and retrieval rankings of two emulators differ by a change of measure with density proportional to t^2/q^4 (Proposition 5), and on the EMIT table the convex stack’s net excess of squared retrieval error lies in the tenth of the entries where that density is smallest.

Training on the retrieval error is regression under a covariate shift whose likelihood ratio is known in closed form and is unbounded as $t \rightarrow 0$; a floor on t truncates it as in the reweighted estimators of [61]. The rate that a fit certified in the weighted norm guarantees is sharp (Propositions 6 and 7), and the price of the weighting can be computed before any training: the effective sample size of the floored weights is of the order of the number of entries below the floor (Proposition 8), and a fraction $(2 - \beta)/2$ of their weight lies below the floor (Corollary 9). On the EMIT table that weight sits in about thirty of the 285 bands, those of the water-vapour absorption, and it keeps most of the states. Networks trained on the weighted objective show the price from both sides (Section 8.1). On the EMIT table the floored weights at a tenth of the median transmission lower the tail of the retrieval error on the physical domain on every split, by nearly a quarter to more than a third on average, and leave the radiance error unchanged, and a rule fixed before the runs, which chooses the floor on the validation block, picks that floor on every split. At lower floors, which keep fewer effective entries, the weighted objective still falls but the radiance error grows, and at the lowest floor every tail with it; the objective at a low floor is attained best on held-out states by the fit one decade above it. Flattening the weights, $\bar{w}_\tau^\kappa$ with $0 < \kappa < 1$ [84], does better than any floor: at a hundredth of the median transmission and $\kappa = 1/2$ the network lowers the radiance error by 8% and the four tails by 32 to 44% on average, each on every split, and it attains the floored objectives on held-out states better than the networks fitted on them; only the albedo error grows, by less than half. Kernel ridge regression fitted on the floored weights, the truncated reweighting of [61], repeats the pattern of the floors (Section 8.3): at a tenth of the median transmission it lowers every physical tail on every split, by 9 to 15%, at 12% more radiance error, and at a thousandth its objective on held-out states falls to 0.08 of the plain fit’s while the radiance error and two of the tails rise. The table holds two failed evaluations of the code, with zero transmission in most bands. A floored weight gives such an entry the weight of u^{-2} ordinary ones, and with the two in the table every weighted network is worse on held-out states than the plain one, on nearly every split even on the weighted objective itself. On a libRadtran table, where the floored weights fall on one band of thirteen, no floor improves on the plain network in both the radiance error and the tails, and the same rule picks one that is worse in the radiance error and in the physical tails. On the published benchmark of [65], the same corpus split by its release into 35,000 training states, with the 6S coefficients as an input and the band as one, the flattened and floored weights lower the radiance error by factors of 3.6 to 4.8 and every retrieval tail by factors of 5.3 to 6.7 on every seed; in the release’s own metrics they cut the mean absolute error to 0.37 to 0.45 of its best model’s and raise the root-mean-square error by about a quarter, all of it from the spherical albedo (Section 8.2).

Choosing a few numbers on the retrieval error is a different matter. The price of choosing among K candidates on a weighted criterion is set by the effective number of states rather than of entries (Proposition 11), and the retrieval weights keep three quarters of the validation states or more. Refitting the weights of the convex stack on the first-order retrieval error, with the members fitted on the forward error, lowers the 95th percentile of the retrieval error on the physical domain below that of the component stack on 39 of 40 held-out halves of the EMIT table, by a quarter at the median, to the level of the kernel on learned features, the best single member (Section 8.4); every stack that no other improves on both counts minimizes one of a one-parameter family of mixed objectives (Proposition 12). Chosen on the cut criterion, the input length scales of a kernel regression move toward the water-vapour column and nearly halve its tail on the physical domain,

at four and a half times its radiance error; at that radiance error, about the networks’ own, the tail is a fifth of the plain network’s (Section 8.5). The stopping epoch of a network, a choice among many close candidates, has little to give either way (Section 8.6).

Outline. Section 2 reviews related work. Section 3 gives the retrieval error and its metric, Section 4 the change of measure, Section 5 what a weighted fit guarantees and what it costs, and Section 6 the price of choosing on a weighted criterion and the family of stacking objectives. Section 7 describes the tables, the emulators and the scores, and Section 8 the experiments.

2 Related work

Radiative transfer and its inversion. The relation $L = a + \rho t / (1 - \rho s)$ for a Lambertian surface goes back to the modelling of satellite measurements of ground reflectance in [28]. Operational atmospheric correction evaluates it with a radiative-transfer code, 6S and its vector version [101, 51], libRadtran [64, 33] or MODTRAN [7], directly or through lookup tables, as in ATCOR [76] and in the algorithms reviewed in [34]. For imaging spectroscopy the retrieval is posed as optimal estimation [78]: ISOFIT inverts the surface and the atmosphere jointly for EMIT [96], and the uncertainty of the retrieved reflectance is quantified in [97]. The carbon-dioxide retrievals of OCO-2 invert a full-physics forward model in the same framework [17, 25, 20, 29], and fast principal-component approximations of the radiative transfer [50] have been judged by the errors they cause in these retrievals [87, 88]. The benchmark of Section 8.2 uses the thirteen bands of Sentinel-2 [27].

Emulators of radiative transfer. Networks have replaced parts of atmospheric models since the emulation of longwave radiation in a climate model [53], and more recently for gas optics and for the whole transfer calculation [99, 98]. Learned parameterizations of subgrid processes [74] are fitted offline and used inside the model, and accuracy offline does not guarantee a stable coupled simulation [9], a counterpart of the gap between forward and retrieval error studied here. In remote sensing, statistical emulators of radiative-transfer models are surveyed in [103], packaged in the toolbox of [77], used for global sensitivity analysis [102] and built on Gaussian processes [67, 13]. The sRTMnet emulator corrects a fast approximation toward MODTRAN inside ISOFIT [10]; Vicent Servera et al. [81] compare regression emulators of the MODTRAN transfer functions and evaluate the corrected reflectance; Lamminpää et al. [1] fit a Gaussian-process emulator of the OCO-2 forward model by cross-validated forward error and use its analytic Jacobians in the retrieval. Surrogates of radiative transfer also serve inversions of exoplanet atmospheres [41], atmospheric sounding [82] and, for atmospheric transport, greenhouse-gas flux inversions [24]. The correction of a cheap code toward an expensive one is multi-fidelity modelling [46, 71, 70] in the tradition of the design and analysis of computer experiments [79]; the pKANrtm emulator of [65] fits the libRadtran residual of 6S with a Kolmogorov–Arnold network [60, 49]. The emulators compared in [85] and used here are networks, Matérn kernel regressions [90, 73] and their combinations. None of these fits is weighted by the retrieval.

Surrogates for inverse problems. In Bayesian inverse problems [44, 92] the effect of a surrogate on the solution has been treated as an approximation error in the likelihood [45], bounded through moments of the emulator error in the Hellinger distance between posteriors [93], and controlled by an L^2 error weighted by the posterior, which suggests where to place training points [40]. Surrogates refined where the posterior concentrates [57, 16, 35], corrected by the residual of the forward model [14], enriched with gradient information [80], or fitted on a reduced parameter space informed by

the likelihood [21, 22, 106] all aim the fit at the inverse problem; the propagation of surrogate uncertainty into the posterior is studied in [75], and training data chosen to match the information needed downstream in [54]. Goal-oriented model reduction [11], goal-oriented inference [58, 59, 89] and goal-oriented surrogates for stochastic inversion [63] fit the reduced model to the quantity of interest. Derivative-informed training fits a network to the sensitivities as well as the values [23, 6], and derivative-informed neural operators [55, 56] accelerate Bayesian inversion [15]; [5] constrain the sensitivities of a neural operator for forward and inverse modelling. The weight here comes from the retrieval equation in closed form and acts on the three component errors jointly through a metric.

Training for the task. Fitting a model by the loss of the decision it feeds is decision-focused learning [26, 104, 62], with the predict-then-optimize loss of [31], its generalization bounds [3], learned surrogate losses [83] and decision-focused surrogate models in process engineering [38]. The retrieval is a decision of this kind, but its sensitivity to the emulator is known in closed form, so the task loss is a weighted forward loss and needs no differentiation through a solver.

Covariate shift and weights. Weighting by a known likelihood ratio is the classical answer to covariate shift [84, 95, 66], with importance-weighted cross-validation [94], estimated ratios [42] and a learning theory of the correction [18, 19]. It is needed when the model is misspecified and not when it is correctly specified [84]; for nonparametric models that are misspecified it is again needed for the best approximation under the target distribution [36], and the benefit of labels from the target depends on how singular the shift is [52]. For kernel ridge regression with an unbounded ratio of finite second moment the truncated reweighting of [61] is minimax optimal, and in general the estimation error of an importance-weighted fit is controlled by the second moment of the weights [19]. For overparameterized networks the effect of importance weights fades as training proceeds without regularization [12]. Truncating weights [43] and smoothing their tail [100] are the importance-sampling counterparts of the floor and of the flattening used here, and the effective sample size of a weighted average is Kish’s [48], discussed at length in [32].

Choosing models and combining them. The kernel parameters of Section 8.5 are chosen on a criterion evaluated on held-out states; cross-validation [91, 2] and its closed form for ridge regression [37] are the usual criteria, and kernel flows [68, 39] learn kernel parameters by how much the kernel interpolant changes when half of the data are dropped. Stacking [105, 8] descends from the combination of forecasts [4]. The networks are trained with Adam [47] and stopped on held-out data [72], in PyTorch [69].

3 The retrieval error of an emulator

An emulator predicts $(\hat{a}, \hat{t}, \hat{s}) = (a + e_a, t + e_t, s + e_s)$, and the retrieval inverts the predicted relation at the true radiance,

$$\hat{\rho} = \frac{L - \hat{a}}{\hat{t} + \hat{s}(L - \hat{a})}, \quad L = L(\rho).$$

Write $q = 1 - \rho s$, so that $\partial_\rho L = t/q^2$.

Lemma 1 (exact form of the retrieval error). *Put*

$$\ell = -\frac{q}{t}(q e_a + \rho e_t) - \rho^2 e_s, \quad \delta = \frac{q}{t}(e_t - s e_a) + \rho e_s - \frac{q}{t} e_a e_s.$$

Whenever the predicted denominator is nonzero,

$$\hat{\rho} - \rho = \frac{\ell + (\rho q/t) e_a e_s}{1 + \delta}.$$

Proof. Let $u = L - a = \rho t/q$ and $D = t + su = t/q$, so that $\rho = u/D$. The predicted denominator is $\hat{D} = \hat{t} + \hat{s}(u - e_a) = D + e_t + u e_s - s e_a - e_a e_s$, and

$$\hat{\rho} - \rho = \frac{u - e_a}{\hat{D}} - \frac{u}{D} = \frac{-e_a D - u(e_t + u e_s - s e_a) + u e_a e_s}{D \hat{D}}.$$

Since $D - su = t$, the numerator equals $-t e_a - u e_t - u^2 e_s + u e_a e_s$. Divide numerator and denominator by $D^2 = t^2/q^2$ and use $u/D = \rho$, $u q^2/t^2 = \rho q/t$ and $\hat{D}/D = 1 + \delta$. $\square$

Corollary 2 (first order). *As the component errors tend to zero relative to t ,*

$$\hat{\rho} - \rho = \ell + O\left(\frac{|e|^2}{t^2}\right), \quad \ell = -\frac{q^2}{t}(\hat{L}(\rho) - L(\rho)) + O\left(\frac{|e|^2}{t^2}\right),$$

where $\hat{L}(\rho) = \hat{a} + \rho \hat{t}/(1 - \rho \hat{s})$ and $|e| = |e_a| + |e_t| + t|e_s|$. The retrieval error is the forward error at ρ divided by the slope $\partial_\rho L = t/q^2$.

Proof. The first statement is Lemma 1 with $|\delta| = O(|e|/t)$. For the second, the forward error is $\hat{L}(\rho) - L(\rho) = e_a + \rho e_t/q + \rho^2 t e_s/q^2 + O(|e|^2)$, and multiplying by $-q^2/t$ gives ℓ . $\square$

Let ν be a probability measure on the reflectances of interest in $(0, R]$, and write $e = (e_a, e_t, e_s)^\top$ and

$$v_\rho = \left(\frac{q_\rho^2}{t}, \frac{\rho q_\rho}{t}, \rho^2 \right)^\top, \quad M(t, s) = \int v_\rho v_\rho^\top \nu(d\rho), \quad q_\rho = 1 - \rho s.$$

By Corollary 2, $\int (\hat{\rho} - \rho)^2 \nu(d\rho) = e^\top M(t, s) e + O(|e|^3/t^3)$ at every entry, so the first-order mean squared retrieval error of an emulator over an evaluation distribution is the mean of a Mahalanobis norm of its component errors, with a metric that depends on the state and the band only through (t, s) .

Proposition 3 (rank of the metric). *If ν is a point mass, M has rank one. If the support of ν contains at least three points, M is positive definite for every $t > 0$ and $s \in [0, 1)$.*

Proof. A point mass gives $M = v_\rho v_\rho^\top$. In general $x^\top M x = \int (v_\rho^\top x)^2 \nu(d\rho)$, and $v_\rho^\top x = p_x(\rho)$ with $p_x(\rho) = x_1(1 - 2s\rho + s^2\rho^2)/t + x_2(\rho - s\rho^2)/t + x_3\rho^2$, a polynomial of degree at most two. Its coefficients in the basis $(1, \rho, \rho^2)$ are $(x_1/t, (x_2 - 2sx_1)/t, (s^2x_1 - sx_2)/t + x_3)$, a triangular invertible map of x , so $p_x \equiv 0$ only for $x = 0$. A nonzero polynomial of degree at most two vanishes at no more than two points, hence $x^\top M x > 0$ for $x \neq 0$ once ν charges three. $\square$

Remark 4. At a single reflectance the retrieval sees one combination of the three component errors, $v_\rho^\top e$, and is blind to the two directions orthogonal to v_ρ . For small t the first two coordinates of v_ρ scale as $1/t$ and the third does not, so path-radiance and transmission errors are weighted by $1/t^2$ and the albedo error by a constant, in agreement with the transfer inequality $|\bar{\rho} - \rho| \leq e_R/t$, $e_R = |e_a| + R|e_t| + R^2 t |e_s|$, of [85], where the albedo enters as $R^2 t |e_s|/t = R^2 |e_s|$.

A comparison with the exact inverse on 10^5 random states, with component errors of the order of the components themselves, reproduces Lemma 1 to within 1.5×10^{-12} .

4 Forward and retrieval rankings differ by a change of measure

Fix ρ and write $d_f = v_\rho^\top e_f$ for an emulator f with component errors e_f . By Corollary 2, to first order the retrieval error is $\hat{\rho}_f - \rho = -d_f$ and the forward error is $r_f = \hat{L}_f(\rho) - L(\rho) = (t/q^2) d_f$. Let p be a distribution over the entries, a state and a band, with $\mathbb{E}_p[d_f^2] < \infty$, and put $g = t^2/q^4$.

Proposition 5 (change of measure). *If $0 < \mathbb{E}_p[g] < \infty$, then $\mathbb{E}_p[r_f^2] = \mathbb{E}_p[g] \mathbb{E}_{p_g}[d_f^2]$ with $dp_g/dp = g/\mathbb{E}_p[g]$. Two emulators f, h are ordered by the forward squared error as they are ordered by the squared retrieval error under p_g . The orders under p and p_g disagree, $\mathbb{E}_p[r_f^2] < \mathbb{E}_p[r_h^2]$ while $\mathbb{E}_p[d_f^2] > \mathbb{E}_p[d_h^2]$, exactly when $\Delta = d_f^2 - d_h^2$ satisfies*

$$\mathbb{E}_p[\Delta] > 0 \quad \text{and} \quad \text{Cov}_p(g, \Delta) < -\mathbb{E}_p[g] \mathbb{E}_p[\Delta].$$

Proof. The first identity is $r_f^2 = g d_f^2$ integrated against p . The second statement follows from $\mathbb{E}_p[g\Delta] = \mathbb{E}_p[g]\mathbb{E}_p[\Delta] + \text{Cov}_p(g, \Delta)$. $\square$

Relative to the retrieval, the forward metric gives an entry of transmission t a weight of order t^2 . On the EMIT table, where the transmissions of the physical domain range over more than twenty orders of magnitude, the forward metric is blind to the entries that set the tail of the retrieval error, and an emulator whose excess squared error sits at small transmission can win on the forward metric and lose on the retrieval. For the convex stack and the kernel on learned features of [85], on three splits at two network widths and on the physical domain above $10^{-3}T_{\text{train}}$, the first-order forward error at $\rho = 0.7$ favors the stack in all six runs, while the first-order retrieval error favors the kernel on features by factors of 5.0 to 9.9 and the exact constrained retrieval by factors of 3.3 to 4.5. The identity of Proposition 5 closes to rounding, and all of the stack's net excess $\sum \Delta$ lies in the tenth of the entries with the smallest g . The Pearson correlation of g and Δ is only -0.014 , because Δ is heavy-tailed; the covariance, not the correlation, is the quantity the proposition needs.

5 What a fit for the retrieval guarantees and what it costs

Consider one output coordinate $c \in \{a, t, s\}$ at one band as a scalar regression on the state x , with training distribution p . The diagonal of the first-order retrieval risk, $\mathbb{E}_p[M_{cc}(x) e_c(x)^2]$, is $\mathbb{E}_p[M_{cc}]$ times the mean squared error under the distribution p_c with

$$\frac{dp_c}{dp}(x) = w_c(x) = \frac{M_{cc}(x)}{\mathbb{E}_p[M_{cc}]}.$$

Fitting for the retrieval is therefore regression under covariate shift, with a likelihood ratio that is known in closed form rather than estimated; the off-diagonal terms of M couple the coordinates but do not change this. The ratio is unbounded: $M_{aa} = \int q_\rho^4 \nu(d\rho)/t^2$ and $M_{tt} = \int \rho^2 q_\rho^2 \nu(d\rho)/t^2$, so $\mathbb{E}_p[w_c^2] < \infty$ requires $\mathbb{E}_p[t^{-4}] < \infty$, which fails for a table whose transmissions come arbitrarily close to zero. Replacing t by $\max\{t, \tau\}$ truncates the ratio at a level of order τ^{-2} . For kernel ridge regression, the unweighted estimator with a suitable regularization is minimax optimal up to a logarithmic factor when the ratio is uniformly bounded, and when it is unbounded with a finite second moment the reweighting by the ratio truncated at a suitable level attains the minimax rate [61]. A retrieval loss with a floor τ is that truncated reweighting, with a ratio the physics supplies.

5.1 The rate a weighted fit guarantees

Write $e_R = |e_a| + R|e_t| + R^2t|e_s|$ and $\bar{\rho}$ for the retrieval constrained to $[0, R]$, which satisfies $|\bar{\rho} - \rho| \leq \min\{R, e_R/t\}$ at every entry [85].

Proposition 6 (the rate a weighted certificate buys). *Let $0 \leq \alpha \leq 1$ and suppose that $\mathbb{E}_p[t^{-2\alpha}e_R^2] \leq \varepsilon^2$.*

1. *If $\alpha = 1$, then $\mathbb{E}_p|\bar{\rho} - \rho|^2 \leq \varepsilon^2$.*
2. *If $\alpha < 1$ and $\mathbb{P}_p(t \leq u) \leq Au^\beta$ for $0 < u \leq u_0$, then for $\varepsilon \leq u_0^{(\beta+2-2\alpha)/2}$,*

$$\mathbb{E}_p|\bar{\rho} - \rho|^2 \leq (R^2A + 1)\varepsilon^{2\beta/(\beta+2-2\alpha)}.$$

Proof. For every $\tau > 0$ the pointwise bound gives $\mathbb{E}_p|\bar{\rho} - \rho|^2 \leq R^2\mathbb{P}_p(t \leq \tau) + \mathbb{E}_p[t^{-2}e_R^2\mathbf{1}\{t > \tau\}] \leq R^2\mathbb{P}_p(t \leq \tau) + \tau^{2\alpha-2}\varepsilon^2$, since $t^{-2} \leq t^{-2\alpha}\tau^{2\alpha-2}$ on $\{t > \tau\}$. At $\alpha = 1$ let $\tau \rightarrow 0$ and use $t > 0$. Otherwise take $\tau = \varepsilon^{2/(\beta+2-2\alpha)} \leq u_0$; then τ^β and $\tau^{2\alpha-2}\varepsilon^2$ both equal $\varepsilon^{2\beta/(\beta+2-2\alpha)}$. $\square$

Proposition 7 (the weighted exponent is attained). *Let $\beta > 0$, $0 \leq \alpha < 1$, $0 < c < r_0 < R$ and $\tau \in (0, 1)$. Take $s = a = 0$, $\rho = r_0$, $\mathbb{P}_p(t \leq u) = u^\beta$ on $(0, 1]$, and the predictor $\hat{t} = t$, $\hat{s} = 0$, $\hat{a} = ct\mathbf{1}\{t \leq \tau\}$. Then $\bar{\rho} - \rho = -c\mathbf{1}\{t \leq \tau\}$,*

$$\begin{aligned} \mathbb{E}_p[t^{-2\alpha}e_R^2] &= \frac{c^2\beta}{\beta+2-2\alpha}\tau^{\beta+2-2\alpha} =: \varepsilon^2, \\ \mathbb{E}_p|\bar{\rho} - \rho|^2 &= c^2\tau^\beta = c^{\frac{4-4\alpha}{\beta+2-2\alpha}}\left(\frac{\beta+2-2\alpha}{\beta}\right)^{\frac{\beta}{\beta+2-2\alpha}}\varepsilon^{\frac{2\beta}{\beta+2-2\alpha}}. \end{aligned}$$

At $\alpha = 1$ the same predictor gives $\mathbb{E}_p|\bar{\rho} - \rho|^2 = \mathbb{E}_p[t^{-2}e_R^2]$, so part 1 of Proposition 6 holds with equality.

Proof. With $s = 0$ the radiance is $L = r_0t$, and the retrieval gives $(L - \hat{a})/\hat{t} = r_0 - c\mathbf{1}\{t \leq \tau\}$, which lies in $[0, R]$. The only error is e_a , hence $e_R = ct\mathbf{1}\{t \leq \tau\}$ and $\mathbb{E}_p[t^{-2\alpha}e_R^2] = c^2 \int_0^\tau u^{2-2\alpha}\beta u^{\beta-1} du$. Solving for τ and substituting into $c^2\tau^\beta$ gives the second identity. $\square$

At $\alpha = 0$ the exponent is the $2\beta/(\beta+2)$ of plain fitting, and it rises continuously to 2 as $\alpha \rightarrow 1$. If the plain fit reaches $\mathbb{E}_p[e_R^2]^{1/2} = O(n^{-r_0})$ and the fit certified at α reaches $\mathbb{E}_p[t^{-2\alpha}e_R^2]^{1/2} = O(n^{-r_\alpha})$, weighting gives the better guarantee exactly when $r_\alpha > r_0(\beta+2-2\alpha)/(\beta+2)$, a factor close to $1 - \alpha$ for small β ; at the EMIT value $\beta \approx 0.12$ and $\alpha = 1/2$ it is 0.53.

5.2 The price of the weighting

For entry weights w the effective sample size of a weighted fit on n entries is $n\mathbb{E}_p[w]^2/\mathbb{E}_p[w^2]$ [48], and for the weights of the retrieval it is set by the transmission tail.

Proposition 8 (the effective sample size of retrieval weighting). *Let $0 < \beta < 2$ and suppose $F(u) = \mathbb{P}_p(t \leq u)$ satisfies $F(u)/(Au^\beta) \rightarrow 1$ as $u \rightarrow 0$, with $\mathbb{E}_p[t^{-4}\mathbf{1}\{t > u_0\}] < \infty$ for some $u_0 > 0$. For the floored weights $w_\tau = \max\{t, \tau\}^{-2}$ and the cut weights $v_\tau = t^{-2}\mathbf{1}\{t > \tau\}$, as $\tau \rightarrow 0$,*

$$\frac{\mathbb{E}_p[w_\tau]^2}{\mathbb{E}_p[w_\tau^2]} \sim \frac{4-\beta}{(2-\beta)^2}F(\tau), \quad \frac{\mathbb{E}_p[v_\tau]^2}{\mathbb{E}_p[v_\tau^2]} \sim \frac{\beta(4-\beta)}{(2-\beta)^2}F(\tau).$$

Proof. Integrating by parts, $\mathbb{E}_p[t^{-k}\mathbf{1}\{t > \tau\}] = -\tau^{-k}F(\tau) + k \int_{\tau}^{\infty} u^{-k-1}F(u) du$ for $k = 2, 4$. As $\tau \rightarrow 0$ the integral over $[u_0, \infty)$ stays bounded, while over (τ, u_0) it is $\frac{A}{k-\beta}\tau^{\beta-k}(1 + o(1))$ times k , which diverges because $\beta < 2 \leq k$. Hence $\mathbb{E}_p[v_{\tau}] \sim A\frac{\beta}{2-\beta}\tau^{\beta-2}$ and $\mathbb{E}_p[v_{\tau}^2] \sim A\frac{\beta}{4-\beta}\tau^{\beta-4}$, and adding $\tau^{-2}F(\tau)$ and $\tau^{-4}F(\tau)$ gives $\mathbb{E}_p[w_{\tau}] \sim A\frac{2}{2-\beta}\tau^{\beta-2}$ and $\mathbb{E}_p[w_{\tau}^2] \sim A\frac{4}{4-\beta}\tau^{\beta-4}$. $\square$

Corollary 9 (where the floored weight sits). *Under the hypotheses of Proposition 8,*

$$\frac{\mathbb{E}_p[w_{\tau}\mathbf{1}\{t \leq \tau\}]}{\mathbb{E}_p[w_{\tau}]} \rightarrow \frac{2-\beta}{2} \quad (\tau \rightarrow 0).$$

Proof. The numerator is $\tau^{-2}F(\tau) \sim A\tau^{\beta-2}$ and the denominator is $A\frac{2}{2-\beta}\tau^{\beta-2}(1 + o(1))$ by the proof of Proposition 8. $\square$

A fit weighted for the retrieval learns, in effect, from about as many entries as lie below its floor, and most of its weight sits on entries that a retrieval at the same floor does not use. On the EMIT table ($\beta \approx 0.12$; floors uT_{train} with T_{train} the median positive training transmission) the floored weights keep an effective fraction of 0.24, 0.11 and 0.07 of the entries at $u = 10^{-1}, 10^{-2}, 10^{-3}$, and 0.76, 0.90 and 0.92 of their weight lies below the floor, against the limit 0.94. The cut weights put none below the floor and keep an effective fraction of 0.13, 0.024 and 0.011.

The weight is concentrated in bands rather than in states. Write $\omega_i = \sum_b w_{ib}$ for the weight of state i and $\pi_{ib} = w_{ib}/\omega_i$ for its spread over the B bands. Over m states,

$$\frac{n_{\text{eff}}}{n} = \frac{(\sum_i \omega_i)^2}{m \sum_i \omega_i^2} \cdot \frac{\sum_i \omega_i^2}{B \sum_i \omega_i^2 \|\pi_i\|^2},$$

the effective fraction of the states times a weighted harmonic mean of the effective number of bands per state, $1/\|\pi_i\|^2$, divided by B ; the identity is $n_{\text{eff}}/n = (\sum_{ib} w_{ib})^2 / (mB \sum_{ib} w_{ib}^2)$ written out. On the training block of one EMIT split the two factors are 0.73 and 0.095 at $u = 10^{-3}$, and 0.91 and 0.27 at $u = 10^{-1}$: the floored weights keep nearly three quarters of the states and about 27 of the 285 bands, which carry 90% of the weight. The eighteen bands 132–136 and 195–207, inside the water-vapour absorption near 1380 and 1870 nm, have more than half of their entries below $10^{-3}T_{\text{train}}$.

Where the weight sits is not where the value of the objective sits, because the absolute errors are smallest where the transmission is (Figure 1b). At the test residuals of the plain network of one EMIT split the entries below the floor carry 11% of the floored objective for the path radiance and 43% for the transmission at $u = 10^{-1}$, and 82% and 76% at $u = 10^{-3}$.

These are fractions of the weight against the physical squared error. The networks of Section 7 are fitted to outputs standardized band by band, and against their own loss the weight of an entry of the component c is $w_{ib}\sigma_{cb}^2$, with σ_{cb} the standard deviation of band b over the training states. The band variances differ by orders of magnitude. On the EMIT table they cut the entry effective fraction of the floored weights of the path radiance to 0.072, 0.021 and 0.017 at the three floors, a fifth to a quarter of the physical fraction below $u = 10^{-1}$, and the effective number of training states, $(\sum_i \omega_i)^2 / \sum_i \omega_i^2$, to 10,700, 7,200 and 8,900 of 18,883 (medians over ten splits; the direct and diffuse terms keep 5,700 to 10,100). The cut weights keep about 7,000 states of the path radiance at the two lower floors.

A floored weight gives its largest value, τ^{-2} , to every entry below the floor, and so to any evaluation of the forward code that failed and returned no transmission. In a band where the other transmissions are of the order of the median T , such an entry weighs as much as $(T/\tau)^2 = u^{-2}$ ordinary ones, a million at $u = 10^{-3}$, more than the table holds. The cut weights give it none, but

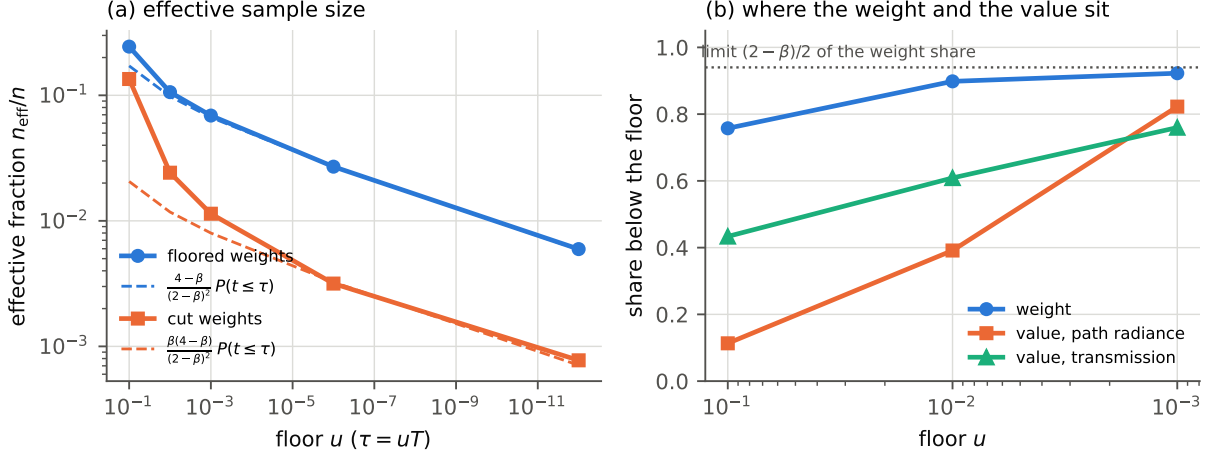


Figure 1: (a) The effective fraction n_{eff}/n of the floored weights $\max\{t, \tau\}^{-2}$ and of the cut weights $t^{-2}\mathbf{1}\{t > \tau\}$ over the entries of the physical domain of the EMIT table, $\tau = uT$ with T the median transmission, and the limits of Proposition 8 at $\beta = 0.12$. (b) On the test block of one EMIT split, the share of the floored weight on the entries below the floor, and the share of the floored objective's value there at the residuals of the plain network, for the path radiance and the transmission; the dotted line is the limit of Corollary 9.

an evaluation need not fail all at once. The EMIT table has two failed evaluations, states 4011 and 7439. Below about 800 nm their transmission and albedo agree with those of their neighbours in the input space. Between 797 and 850 nm their fluxes fall to 10^{-4} of the neighbours' and their spherical albedo rises to 2; from about 1030 nm on both fluxes are exactly zero, in 197 of the 285 bands, and the albedo is at least one in 227 bands, exactly 2 in 107. No other state has an albedo of one or more. In the five bands from 812 to 842 nm their transmission, a fifth down to a two-hundredth of the neighbours', is still above $10^{-2}T$, so the cut weights count these entries too, and their errors are as large as the fluxes themselves. Against the physical squared error they are two states in 23,313 and change none of the fractions above by more than one percent. In the coordinates of training, where the weight of an entry carries the variance of its band, they are the only states below the floor in more than 160 bands from 850 nm on, among them the near-infrared bands, where the path radiance varies more than anywhere outside the visible. On the eight splits that keep both in the training block they hold 16% of the floored weight of the path radiance at $u = 10^{-2}$ and 45% at $u = 10^{-3}$, and the effective number of training states falls to 75 and 10. Section 8.1 fits the networks with and without them.

Mixing the two objectives buys effective sample size back once α is small against $\sqrt{\rho_\tau}$. With $\bar{w}_\tau = w_\tau/\mathbb{E}_p[w_\tau]$ and $\rho_\tau = 1/\mathbb{E}_p[\bar{w}_\tau^2]$, the weights $w = (1 - \alpha) + \alpha\bar{w}_\tau$ have mean one and effective fraction $1/(1 - \alpha^2 + \alpha^2/\rho_\tau)$, while $\mathbb{E}_p[\bar{w}_\tau e_R^2] \leq \mathbb{E}_p[w e_R^2]/\alpha$ keeps the retrieval certificate up to the factor $1/\alpha$. For the path radiance in the coordinates of training at $u = 10^{-2}$, where $\rho_\tau = 0.021$, the fraction is 0.89, 0.35 and 0.08 at $\alpha = 0.05, 0.2$ and 0.5 .

Corollary 10 (rate against effective sample size). *Under the hypotheses of Proposition 8, suppose a plain fit on n entries has $\mathbb{E}_p[e_R^2] \leq Cn^{-\gamma}$ and a fit with the normalized floored weights $\bar{w}_\tau$ has $\mathbb{E}_p[\bar{w}_\tau e_R^2] \leq Cn_{\text{eff}}^{-\gamma}$ with $n_{\text{eff}} = n \mathbb{E}_p[w_\tau]^2/\mathbb{E}_p[w_\tau^2]$, for some $\gamma > 0$. With the floor chosen as a power*

of n ,

$$\mathbb{E}_p|\bar{\rho} - \rho|^2 = O(n^{-\beta\gamma/(2+\beta)}) \text{ (plain)}, \quad \mathbb{E}_p|\bar{\rho} - \rho|^2 = O(n^{-\beta\gamma/(2+\beta\gamma)}) \text{ (floored weights)},$$

so the weighted exponent is the larger exactly when $\gamma < 1$.

Proof. For both fits $\mathbb{E}_p|\bar{\rho} - \rho|^2 \leq R^2 F(\tau) + \mathbb{E}_p[t^{-2}e_R^2 \mathbf{1}\{t > \tau\}]$. For the plain fit the second term is at most $\tau^{-2}Cn^{-\gamma}$, and $\tau = n^{-\gamma/(2+\beta)}$ balances it against $R^2 F(\tau) \sim R^2 A\tau^\beta$. For the weighted fit $t^{-2}\mathbf{1}\{t > \tau\} \leq w_\tau$, so the second term is at most $\mathbb{E}_p[w_\tau] Cn_{\text{eff}}^{-\gamma}$, which by Proposition 8 is of order $\tau^{\beta-2}(n\tau^\beta)^{-\gamma}$; $\tau = n^{-\gamma/(2+\beta\gamma)}$ balances it. $\square$

On EMIT the learning curves of [85] give mean-square exponents of about 0.8 to 1.2 on the first three components, and with $\beta \approx 0.12$ the two exponents agree to within about one percent either way. Rates cannot decide whether weighting pays on this table; the constants decide it, and Section 8.1 measures them.

6 Choosing on the retrieval error

Fitting a model on a weighted objective pays the effective sample size of the entries. Choosing among a few candidates fitted on the forward error pays much less, because the choice is made on a sum over states.

Proposition 11 (choosing on a weighted criterion). *Let $Z_1, \dots, Z_m$ be independent copies of a state Z with B entries, let $v_b(Z) \geq 0$ be entry weights with state weight $\omega(Z) = \sum_b v_b(Z) \leq \omega_{\max}$, and let $\ell_{k,b}(Z) \in [0, L]$ be the entry losses of K candidates. Put $R_k = \mathbb{E} \sum_b v_b \ell_{k,b}$, let $\hat{R}_k$ be its mean over the m states, and let $m_{\text{eff}} = m \mathbb{E}[\omega]^2 / \mathbb{E}[\omega^2]$. With probability at least $1 - \delta$ the candidate $\hat{k}$ that minimizes $\hat{R}_k$ satisfies*

$$R_{\hat{k}} - \min_k R_k \leq 2L \mathbb{E}[\omega] \left(\sqrt{\frac{2 \log(2K/\delta)}{m_{\text{eff}}}} + \frac{2\omega_{\max}}{3 \mathbb{E}[\omega]} \frac{\log(2K/\delta)}{m} \right).$$

Proof. The state loss $X_k = \sum_b v_b \ell_{k,b}$ lies in $[0, L\omega_{\max}]$ and has $\text{Var } X_k \leq \mathbb{E}X_k^2 \leq L^2 \mathbb{E}[\omega^2] = L^2 \mathbb{E}[\omega]^2 m / m_{\text{eff}}$. With $x = \log(2K/\delta)$, Bernstein's inequality gives $|\hat{R}_k - R_k| \leq \varepsilon$ with $\varepsilon = \sqrt{2x \text{Var } X_k / m} + 2L\omega_{\max}x / (3m)$ except on an event of probability at most δ/K , and a union bound over k makes this hold for every k with probability at least $1 - \delta$. Then $R_{\hat{k}} \leq \hat{R}_{\hat{k}} + \varepsilon \leq \hat{R}_{k^*} + \varepsilon \leq R_{k^*} + 2\varepsilon$ for the best candidate k^* , and 2ε is at most the stated bound. $\square$

The weights enter only through the effective number of states and the ratio $\omega_{\max}/\mathbb{E}[\omega]$. On an EMIT validation block of $m = 2,098$ states the cut weights keep 1,962, 1,831 and 1,639 effective states at $u = 10^{-1}, 10^{-2}, 10^{-3}$, with $\omega_{\max}/\mathbb{E}[\omega]$ between 2.6 and 3.0, and the floored weights 1,917, 1,763 and 1,579. For $K = 32$ and $\delta = 0.05$ the bound is 0.18 to 0.20 times $L \mathbb{E}[\omega]$, against 0.17 with no weights at all: weighting the criterion raises the price of choosing by eight to eighteen percent, while it divides the effective data of a weighted fit by fourteen at $u = 10^{-3}$.

A convex stack is a choice of this kind with a continuum of candidates. For members with component errors $e^{(j)}$ and weights w on a product W of simplices, write $F(w)$ and $G(w)$ for the first-order forward and retrieval errors of the stack; both are convex quadratics in w .

Proposition 12 (the mixtures trace the frontier). *Let W be a product of simplices and $F(w) = w^\top Bw$, $G(w) = w^\top Aw$ with A, B positive semidefinite. If $w^* \in W$ is weakly Pareto optimal for (F, G) , that is, no $w \in W$ has $F(w) < F(w^*)$ and $G(w) < G(w^*)$, then w^* minimizes $\alpha F + (1 - \alpha)G$ over W for some $\alpha \in [0, 1]$.*

Proof. The set $S = \{(F(w), G(w)) + u : w \in W, u \in [0, \infty)^2\}$ is convex, since F and G are convex and W is convex. Weak optimality says that $(F(w^*), G(w^*)) - (0, \infty)^2$ does not meet S , so the point lies on the boundary of S , and a supporting line gives $\lambda \neq 0$ with $\lambda \cdot s \geq \lambda \cdot (F(w^*), G(w^*))$ for all $s \in S$. Since S is closed upward, $\lambda \geq 0$; normalize λ to $(\alpha, 1 - \alpha)$. $\square$

Lemma 13 (the path is monotone). *For $0 \leq \alpha_1 < \alpha_2 \leq 1$ let w_i minimize $\alpha_i F + (1 - \alpha_i)G$ over W . Then $F(w_2) \leq F(w_1)$ and $G(w_2) \geq G(w_1)$.*

Proof. Write $x = F(w_1) - F(w_2)$ and $y = G(w_2) - G(w_1)$. Optimality of w_1 and of w_2 gives $\alpha_1 x \leq (1 - \alpha_1)y$ and $(1 - \alpha_2)y \leq \alpha_2 x$. Multiplying the first by $1 - \alpha_2$ and the second by $1 - \alpha_1$ and chaining, $(1 - \alpha_2)\alpha_1 x \leq (1 - \alpha_1)\alpha_2 x$, that is $(\alpha_2 - \alpha_1)x \geq 0$, so $x \geq 0$. Then the first inequality gives $(1 - \alpha_1)y \geq \alpha_1 x \geq 0$, so $y \geq 0$. $\square$

The proof of Proposition 12 uses only that W is convex and that F and G are convex on it, so the same holds for any convex class of predictors, for instance the functions of norm at most r in a reproducing kernel Hilbert space; kernel ridge regression with the entry weights $(1 - \alpha) + \alpha g/\bar{g}$ is the penalized form of that problem.

7 Tables, emulators and scores

EMIT. The first table has 23,313 evaluations of a radiative-transfer code provided by the Jet Propulsion Laboratory as a lookup table for EMIT atmospheric correction. The input is the atmospheric and viewing state in $\mathbb{R}^6$ (aerosol optical depth, surface elevation, water-vapour column, relative azimuth, solar and view zenith angles), and the output is four spectral components on the 285-point EMIT grid from 381 to 2493 nm: path radiance $Y_1 = a$, direct and diffuse fluxes Y_2 and Y_3 with $t = Y_2 + Y_3$, and spherical albedo $Y_4 = s$ [85]. Each split draws a test block of 2,331 states and a validation block of 2,098 from a seeded permutation and fits on the remaining 18,884; the splits are the ten seeds 101–110 of [85]. The two failed evaluations of Section 5.2, states 4011 and 7439, are both in the training block on eight splits; split 104 has one in its test block and split 105 one in its validation block. The networks of Section 8.1 are fitted with them and, removed from whichever block holds them, without them; the members of the stacks of Section 8.4, the kernels of Section 8.5 and the networks of Section 8.6 are fitted on the table as provided.

libRadtran. The second table is the libRadtran half of the corpus of [65] over the thirteen Sentinel-2 bands, restricted to the 9,722 states that carry all thirteen: the path reflectance, total transmittance and spherical albedo for each state and band, with the seven continuous state variables and the one-hot aerosol model and atmosphere profile as inputs [85]. Twelve of the thirteen bands keep a transmission above 0.14 on every state; band B10 at 1372 nm, in a strong water-vapour band, holds every smaller transmission. The splits use the same seeds and proportions (972 test, 7,875 fitting states).

The published libRadtran benchmark. The release of [65] pairs 6S and libRadtran values of the path reflectance a , the total transmittance t and the spherical albedo s for 50,000 states over the thirteen Sentinel-2 bands of [27], with its own state-level split into 35,000 training, 7,500 validation and 7,500 test states and its own code for the metrics. Its files hold 609,722 of the 650,000 rows of one state and one band, the rest removed by its quality control, and we fit and score on the rows present, 426,761, 91,490 and 91,471. Each row is fitted separately and the band enters through the inputs: eight state and band variables and the one-hot aerosol model and atmosphere profile,

to which we add the 6S coefficients of the same row, and the network predicts the libRadtran coefficients minus the 6S ones, the residual form of the release’s best models. The networks have four hidden layers of 384 SiLU units [30] and three outputs, are trained with Adam [47] and a one-cycle schedule [86] for 80 epochs at batch size 1,024, and are kept at the epoch with the lowest validation value of their own objective. The weighted arms use the weights of J_u row by row at the true transmission, $\max\{t, \tau\}^{-2}$, $R^2 \max\{t, \tau\}^{-2}$ and $R^4 t^2 \max\{t, \tau\}^{-2}$ for the three coefficients with $\tau = uT_{\text{train}}$, each normalized to mean one on the training rows. The forward scores are the release’s: the root-mean-square, mean absolute and symmetric mean absolute percentage errors over the three coefficients of every test row, and R^2 averaged over the coefficients. The retrieval scores are those below, with the top-of-atmosphere reflectance $a + \rho t / (1 - \rho s)$ in the role of the radiance.

Emulators. The members are the six families of [85]: a cubic ridge regression, Matérn kernel ridge regression on the inputs with an isotropic and with an input-scaled metric, a network of three hidden layers of 512 units, the network with a kernel correction of its residuals, and kernel ridge regression on the network’s last hidden features. The kernel regressions are fitted on the full fitting block in the space of the leading 64 principal components of each standardized component, with their hyperparameters chosen on the validation block; the network is trained with AdamW and a cosine schedule for up to 150 epochs and stopped on the validation loss with a patience of 25 epochs. The convex stacks of [105, 8] combine the six members with weights on a simplex for each component.

Scores. All scores are on the test block, at $\rho = 0.7$: the mean relative ℓ^2 error of the radiance across bands, and the 95th percentile of the retrieval error $|\hat{\rho} - \rho|$, in points of reflectance (100 times the absolute error), over all entries and over the physical domain above fT_{train} for $f = 10^{-12}, 10^{-3}, 10^{-2}$, that is, the entries with $t \geq fT_{\text{train}}$, $0 \leq s < 1$ and $1 - \rho s \geq 0.3$, where a failed inversion is counted as a failure and left out of the percentile [85].

8 Experiments

8.1 Networks fitted on the retrieval error

The network of Section 7 is trained on each component with the weights of the first-order bound,

$$J_u = \mathbb{E} \left[\frac{e_a^2 + R^2(e_{Y_2}^2 + e_{Y_3}^2) + R^4 t^2 e_s^2}{\max\{t, uT_{\text{train}}\}^2} \right], \quad R = 0.9,$$

normalized to mean one on the training block of each component and stopped on the same weighted loss of the validation block (the *floored* arms). Three variants change one thing each: the *floored* weights at $u = 10^{-2}$ stopped on the plain validation loss, the mixed weights $(1 - \alpha) + \alpha \bar{w}_\tau$ of Section 5.2 with $\alpha = 0.2$ at $u = 10^{-2}$, and the cut weights v_τ of Proposition 8 at $u = 10^{-2}$, normalized and used for stopping in the same way. The *flattened* arms raise the *floored* weights to a power $\kappa \in (0, 1)$, the device Shimodaira proposed for covariate shift [84], and normalize them to mean one in the same way: at $u = 10^{-2}$ with $\kappa = 1/4$ and $\kappa = 1/2$, and with $\kappa = 1/4$ stopped on the plain validation loss at $u = 10^{-2}$ and 10^{-3} . For the path radiance and the two fluxes the flattened weights at κ are, above the floor, those of the certificate of Proposition 6 at $\alpha = \kappa$; the albedo’s become constant above the floor. We write K_u for the objective J_u with v_τ in place of the *floored* weights. The *joint* arms train one network for all four components on the first-order

arm	radiance [pp]	all bands	phys. 10^{-12}	phys. 10^{-3}	phys. 10^{-2}	failed [pp]
<i>without the two failed evaluations</i>						
floored, $u = 10^{-1}$	-0.01 ± 0.01 (9)	-8.7 ± 2.4 (10)	-2.3 ± 1.3 (10)	-1.3 ± 0.2 (10)	-0.7 ± 0.1 (10)	-0.07
floored, $u = 10^{-2}$	$+0.26 \pm 0.02$ (0)	-19.9 ± 2.6 (10)	-2.2 ± 1.2 (10)	-0.1 ± 0.2 (7)	$+0.6 \pm 0.1$ (0)	-0.10
floored, $u = 10^{-3}$	$+1.33 \pm 0.05$ (0)	$+3.1 \pm 1.9$ (0)	$+26.9 \pm 2.3$ (0)	$+15.2 \pm 1.0$ (0)	$+13.1 \pm 0.7$ (0)	-0.10
floored, $u = 10^{-2}$, plain stopping	$+0.26 \pm 0.01$ (0)	-19.9 ± 2.6 (10)	-2.2 ± 1.3 (10)	-0.1 ± 0.2 (6)	$+0.6 \pm 0.1$ (0)	-0.10
mixed, $\alpha = 0.2$, $u = 10^{-2}$	-0.00 ± 0.01 (5)	-6.5 ± 2.0 (10)	-1.4 ± 1.1 (9)	-0.4 ± 0.2 (10)	-0.1 ± 0.1 (8)	-0.08
flattened, $\kappa = 0.25$, $u = 10^{-2}$	-0.03 ± 0.01 (10)	-12.3 ± 2.6 (10)	-3.0 ± 1.4 (10)	-1.0 ± 0.2 (10)	-0.5 ± 0.1 (10)	-0.08
flattened, $\kappa = 0.5$, $u = 10^{-2}$	-0.03 ± 0.01 (10)	-24.8 ± 3.0 (10)	-4.3 ± 1.3 (10)	-1.5 ± 0.2 (10)	-0.7 ± 0.1 (10)	-0.10
flattened, $\kappa = 0.25$, $u = 10^{-2}$, plain stopping	-0.02 ± 0.01 (10)	-12.4 ± 3.0 (10)	-2.6 ± 1.7 (9)	-1.0 ± 0.2 (10)	-0.5 ± 0.1 (10)	-0.08
flattened, $\kappa = 0.25$, $u = 10^{-3}$, plain stopping	-0.02 ± 0.01 (10)	-23.4 ± 3.5 (10)	-3.0 ± 1.3 (10)	-1.0 ± 0.2 (10)	-0.4 ± 0.1 (10)	-0.09
cut, $u = 10^{-2}$	$+0.32 \pm 0.02$ (0)	$+5.6 \pm 1.8$ (0)	-6.5 ± 1.4 (10)	-0.8 ± 0.2 (10)	-0.0 ± 0.1 (6)	+0.98
joint, $u = 10^{-2}$	$+0.17 \pm 0.02$ (0)	-27.1 ± 1.8 (10)	-2.4 ± 1.4 (9)	-0.1 ± 0.2 (6)	$+0.5 \pm 0.1$ (0)	-0.10
joint, $u = 10^{-3}$	$+0.83 \pm 0.03$ (0)	-8.1 ± 2.9 (10)	$+23.6 \pm 2.0$ (0)	$+13.7 \pm 1.2$ (0)	$+11.8 \pm 0.9$ (0)	-0.10
<i>the table as provided</i>						
floored, $u = 10^{-1}$	$+0.55 \pm 0.16$ (0)	$+0.9 \pm 2.6$ (4)	$+2.2 \pm 1.6$ (2)	$+1.5 \pm 0.5$ (0)	$+1.3 \pm 0.4$ (0)	+0.05
floored, $u = 10^{-2}$	$+5.56 \pm 5.08$ (0)	$+18.6 \pm 45.5$ (1)	$+45.5 \pm 53.7$ (0)	$+38.9 \pm 59.5$ (0)	$+37.9 \pm 60.6$ (0)	+0.02
floored, $u = 10^{-3}$	$+18.55 \pm 1.77$ (0)	$+350.8 \pm 142.8$ (0)	$+397.8 \pm 137.0$ (0)	$+413.1 \pm 140.8$ (0)	$+403.6 \pm 144.8$ (0)	+0.15
joint, $u = 10^{-2}$	$+0.85 \pm 0.24$ (0)	-20.1 ± 5.3 (10)	$+2.7 \pm 4.5$ (3)	$+2.5 \pm 1.0$ (0)	$+2.7 \pm 0.8$ (0)	-0.09
joint, $u = 10^{-3}$	$+6.26 \pm 3.25$ (0)	$+28.0 \pm 56.1$ (0)	$+70.6 \pm 55.9$ (0)	$+67.4 \pm 59.8$ (0)	$+63.5 \pm 54.6$ (0)	-0.04

Table 1: Networks fitted on the retrieval error, EMIT: difference from the plain network of the same run in the radiance error and in the 95th percentiles of the retrieval error (points), mean $\pm$ standard deviation over splits, with the number of splits on which the arm is lower in parentheses, and the mean difference in the share of failed inversions on the physical domain above $10^{-3}T_{\text{train}}$. Floored: weights $\bar{w}_\tau$; mixed: $(1 - \alpha) + \alpha\bar{w}_\tau$; flattened: $\bar{w}_\tau^\kappa$ renormalized; cut: v_τ ; joint: one network for all four components. Upper block: the table without its two failed evaluations, splits 101–110; lower block: the table as provided, splits 101–110.

retrieval error at $\rho \in \{0.1, 0.4, 0.7\}$ with t floored at uT_{train} , plus 10^{-2} times the standardized squared error. Every run also fits the plain network on the same split with the same seed. Training is deterministic: on the table as provided the plain networks of the different runs agree to four decimals of the radiance error on every split, so every difference below is paired. Run on the table as provided, the code behind the upper block of Table 1 reproduces the runs of its lower block on splits 101 and 105, every arm to four decimals of every score on the same machine, so the two blocks differ in the two failed evaluations alone. Table 1 gives the differences on the EMIT table without its two failed evaluations and, in its lower block, on the table as provided. Without the failed evaluations the plain network has a radiance error of 0.350%, an all-band tail of 64.9 points and physical tails of 9.7, 3.5 and 2.1 points at the three floors.

Without the failed evaluations the floored weights at $u = 10^{-1}$ improve on the plain network in every tail on every split. The tails on the physical domain fall by nearly a quarter at $10^{-12}T_{\text{train}}$ and by more than a third at 10^{-3} and 10^{-2} , the all-band tail falls by 8.7 points, the share of failed inversions above $10^{-3}T_{\text{train}}$ falls from 0.10% to 0.04%, and the radiance error falls by 0.01 percentage points, on nine splits of ten. On the test blocks the network has the lower value of the floored objective at all three floors on every split, by median factors of 0.64, 0.42 and 0.56 at $u = 10^{-1}, 10^{-2}, 10^{-3}$, and of K_u at 10^{-2} and 10^{-3} on every split (0.36 and 0.42). The components move apart: the network lowers the error of the path radiance by a quarter and that of the direct flux by nearly a tenth, and it triples that of the albedo, from 1.6% to 4.8%. The albedo weights of J_u , $R^4 t^2 / \max\{t, \tau\}^2$, vanish with the transmission, and so does the fit: on split 101 the network's albedo error is 0.64% against 0.70% on the entries above $10^{-1}T_{\text{train}}$, and 11.5% against 3.8% on those between 10^{-2} and $10^{-1}T_{\text{train}}$.

At the lower floors the objective and the tails part. The floored network at $u = 10^{-2}$ has the lower value of its objective on held-out states on every split (median factor 0.84) and trades one error for another: its all-band tail falls by 20 points and its tail above $10^{-12}T_{\text{train}}$ by 2.2, while its

radiance error rises by 0.26 percentage points and its tail above $10^{-2}T_{\text{train}}$ by 0.55. At $u = 10^{-3}$ the objective is again lower on every split (0.54), and every error rises, the radiance error by 1.3 points and the physical tails by 13 to 27. The joint arms, one network for all four components, trade the same way and a little more cheaply: at $u = 10^{-2}$ the all-band tail falls by 27 points on every split for a rise of 0.17 percentage points in the radiance error, and at 10^{-3} the physical tails rise by 12 to 24 points. The objective is a mean, and at $u = 10^{-3}$ most of its value sits on the entries below the floor (Section 5.2); a network can lower it there while the 95th percentile, which the rest of the physical domain sets, grows. Nor are the lower floors attained best by the networks fitted on them. On held-out states the lowest $J_{10^{-2}}$ belongs to the network fitted at $u = 10^{-1}$ on nine splits of ten and to the mixed weights on the tenth, never to the network fitted on it (median factor 0.42 against 0.84), and the lowest $J_{10^{-3}}$ to a network fitted at $u = 10^{-2}$ on every split (0.15 against 0.54). The cut weights make the point most plainly: the network fitted on $K_{10^{-2}}$ has a higher value of it on held-out states than the plain network on nine splits of ten (median factor 1.21), against 0.36 for the floored network at $u = 10^{-1}$. In the coordinates of training the floored weights at $u = 10^{-1}$ keep an effective fraction of 0.046 to 0.072 of the entries of the first three components, against 0.016 to 0.024 at $u = 10^{-2}$ and 0.007 to 0.013 for the cut weights there, and at the size of this table the entries a higher floor keeps are worth more than a closer match between the objective fitted and the objective scored.

Stopping on the plain validation loss moves none of the mean differences at $u = 10^{-2}$ by more than 0.07 points (on a single split by at most 1.4 points, in the all-band tail), so the trade comes from the training weights and not from the stopping rule. The mixture with $\alpha = 0.2$ lowers the tails on average at an unchanged radiance error, by less than the floored weights at $u = 10^{-1}$ on each of the five errors of the table. The cut weights appear to lower the tail above $10^{-12}T_{\text{train}}$ by 6.5 points, but a failed inversion is left out of the percentile. The cut network fails on 1.1% of the entries above $10^{-3}T_{\text{train}}$, against 0.1% for the plain network, and on more than 5% of those above $10^{-12}T_{\text{train}}$ on every split, so that its tail there is infinite once the failures are counted. Counting every failure as an infinite error, on split 101 the tail above $10^{-3}T_{\text{train}}$ is 3.26 points for the cut network, 3.23 for the plain one and 2.19 for the floored network at $u = 10^{-1}$, and 3.8%, 3.5% and 2.6% of the entries have an error above five points.

Flattening the weights does better than any floor. At $u = 10^{-2}$ and $\kappa = 1/2$, weights proportional to $1/\max\{t, \tau\}$ for the path radiance and the fluxes, the network lowers the radiance error on every split, from 0.350% to 0.322% on average (8%, split by split 5 to 10%), and every tail on every split: the all-band tail from 64.9 to 40.1 points and the physical tails from 9.7, 3.5 and 2.1 to 5.3, 2.0 and 1.4 points, by 44, 42 and 32% on average (split by split 28–57%, 35–44% and 26–37%). The share of failed inversions above $10^{-3}T_{\text{train}}$ falls from 0.10% to 0.005%. Against the floored network at $u = 10^{-1}$ it is lower in the radiance error, the all-band tail and the tails above 10^{-12} and $10^{-3}T_{\text{train}}$ on every split, and higher in the tail above $10^{-2}T_{\text{train}}$ on nine splits of ten (1.38 against 1.32 points on average). It lowers the errors of the path radiance, the direct flux and the diffuse flux on every split (medians 0.49% to 0.37%, 0.95% to 0.86% and 0.39% to 0.38%) and raises the albedo’s on every split, from 1.56% to 2.24%, against a tripling at $u = 10^{-1}$. It also attains the weighted objectives on held-out states better than the networks fitted on them: of all the networks fitted on one component at a time it has the lowest value of $J_{10^{-2}}$, $K_{10^{-2}}$ and $K_{10^{-3}}$ on every split, of $J_{10^{-1}}$ on nine and of $J_{10^{-3}}$ on six, with median factors against the plain network of 0.19 for $J_{10^{-2}}$, where the network fitted on $J_{10^{-2}}$ reaches 0.84, and of 0.54 for $J_{10^{-1}}$, where the network fitted on it reaches 0.64. The weights with $\kappa = 1/4$ give the same radiance error and most of the gain in the tails (30, 30 and 25% on the physical domain), and stopping them on the plain validation loss moves each of these averages by at most five points. None of this is explained by the effective sample size alone. Over the entries of the table the flattened weights at $u = 10^{-2}$ keep

an effective fraction of 0.37 at $\kappa = 1/4$ and 0.15 at $\kappa = 1/2$, against 0.11 for the floored weights at that floor and 0.24 at $u = 10^{-1}$: the best network keeps fewer effective entries than the floored network at $u = 10^{-1}$. What flattening keeps, and a higher floor gives up, is the order of the weights between 10^{-2} and $10^{-1}T_{\text{train}}$, where a floor at 10^{-1} makes them equal; what it removes is most of their range, a factor of 100 between 10^{-2} and 1 times T_{train} instead of 10^4 .

The floor itself can be chosen on the validation block. A second set of runs trains the network on the floored weights at six floors, $u \in \{10^3, 1, 0.3, 0.1, 0.03, 0.01\}$, on each split, and keeps the one with the lowest cut objective K_{10-2} on the validation block, a rule fixed before the runs. On the EMIT table it picks $u = 10^{-1}$ on every split, the floor whose network also has the lowest tail above $10^{-3}T_{\text{train}}$ on the test block, and so it reproduces the first row of Table 1: the four tails fall by 8.7, 2.3, 1.3 and 0.73 points on every split at an unchanged radiance error. The floors on either side trade in opposite directions. At $u = 1$ and above the floored weights are nearly constant in the transmission and the fit is a least-squares fit in physical units: it lowers the radiance error by 0.10 percentage points on every split, from 0.35% to 0.25%, and raises the tails on average, the one above $10^{-12}T_{\text{train}}$ by 4 to 10 points. Weighting by the band variance alone does the same to within 0.05 points in every score, so the albedo weights play no part there. At $u = 0.03$ the all-band tail falls by 19 points and the radiance error rises by 0.08, the trade of $u = 10^{-2}$ on a smaller scale. The rule settles on the one floor that improves both.

With the failed evaluations (lower block) every floored and joint arm raises the radiance error and the tails above 10^{-3} and $10^{-2}T_{\text{train}}$ on every split, and only the joint arm at $u = 10^{-2}$ has a lower all-band tail on all ten. With the failed states the plain network has the lower value of the weighted objective itself on the test blocks of nine, nine and ten of the ten splits at $u = 10^{-1}, 10^{-2}, 10^{-3}$, by median factors of 4.3, 9.9 and 8.3, while on the training blocks of splits 101–103 the weighted network reaches 0.03 to 0.06 of the plain network’s value: it fits the entries that carry its objective and does not generalize from them. Two effects of the failed states combine. At $u = 10^{-2}$ and 10^{-3} they leave the floored weights of the path radiance 75 and 10 effective training states (Section 5.2). At $u = 10^{-1}$ they hold under one percent of the weight, but no smooth fit returns a zero flux between neighbours with ordinary ones, and their share of the value of the objective is far larger. On split 104, whose test block holds one of them, the ratio of the floored objective of the plain network’s residuals to their plain mean square is 23 for the direct flux and 86 for the diffuse flux at $u = 10^{-1}$, against 0.12 to 0.42 and 0.24 to 0.55 on the other nine splits: that one state carries about 99% of the objective of each flux. The weighted network at $u = 10^{-1}$ loses its accuracy on those two components, with errors of 2.4% and 0.90% on split 101 against 0.86% and 0.40% without the failed states. The split that puts a failed state in the validation block, 105, is the worst of all, since the weighted stopping rule there is decided by that state; at $u = 10^{-2}$ its tail above $10^{-2}T_{\text{train}}$ rises by 208 points, against 7 to 44 on the other splits. The failed states also cost the plain network part of its accuracy. Removing them lowers its radiance error on each of the ten splits, by 0.035 percentage points on average, and its albedo error from 2.93% to 1.57%, while its physical tails move by less than a quarter of a point, up on some splits and down on others.

On the libRadtran table, which has no failed evaluations, the floored and joint arms raise the radiance error on every split, by 0.03 to 0.48 percentage points against a plain error of 0.356%, and the tail above $10^{-3}T_{\text{train}}$ on nine or ten splits of ten. There the floored weights put 86%, 99.8% and essentially all of their mass on band B10 at the three floors, so that the network is asked to fit one band of thirteen, the band where the retrieval is ill-conditioned, at the expense of the other twelve. The rule that chooses the floor on the validation block fails there. It picks $u = 10^{-2}$ on every split, the floor closest to its own criterion, and that network raises the radiance error by 0.06 percentage points and the tails above 10^{-3} and $10^{-2}T_{\text{train}}$ by 0.12 points on every split, and lowers only the all-band tail, by 2.6 points on nine splits. No floor improves on the plain network there in both the

emulator	RMSE	MAE	R^2	SMAPE [%]	RMSE by coefficient		
					ρ_{path}	T	S
pKANrtm [65]	0.00619	0.00186	0.9947	3.91			
sRTMnet [10], trained in [65]	0.00691	0.00211	0.9942	4.28			
DA [65]	0.00784	0.00261	0.9935	4.95			
GPAC [65]	0.00865	0.00301	0.9928	5.58			
FC [65]	0.00948	0.00342	0.9919	6.13			
RF [65]	0.01075	0.00404	0.9906	7.02			
UMR [65]	0.01394	0.00546	0.9869	9.85			
6S as the emulator	0.09790	0.05348	0.6386	32.38	0.02065	0.16183	0.04623
network, plain	0.00669 ± 0.00043	0.00225 ± 0.00029	0.9946 ± 0.0005	4.13 ± 0.30	0.00099 ± 0.00009	0.00666 ± 0.00077	0.00943 ± 0.00042
network, floored, $u = 10^{-1}$	0.00785 ± 0.00005	0.00083 ± 0.00002	0.9899 ± 0.0002	2.13 ± 0.02	0.00041 ± 0.00001	0.00216 ± 0.00007	0.01341 ± 0.00010
network, flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.00792 ± 0.00002	0.00069 ± 0.00001	0.9896 ± 0.0001	1.96 ± 0.03	0.00037 ± 0.00001	0.00170 ± 0.00004	0.01361 ± 0.00004
	radiance [%]	all bands	phys. 10^{-12}	phys. 10^{-3}	phys. 10^{-2}	failed [%]	
6S as the emulator	20.565	75.55	75.55	74.04	73.81	0.000	
network, plain	0.809 ± 0.104	3.21 ± 0.21	2.67 ± 0.24	2.38 ± 0.20	2.31 ± 0.19	0.097 ± 0.025	
network, floored, $u = 10^{-1}$	0.225 ± 0.008	0.52 ± 0.02	0.48 ± 0.02	0.45 ± 0.02	0.44 ± 0.01	0.000 ± 0.001	
network, flattened, $\kappa = 1/2$, $u = 10^{-2}$	0.170 ± 0.004	0.48 ± 0.01	0.44 ± 0.01	0.41 ± 0.01	0.40 ± 0.01	0.000 ± 0.000	

Table 2: The published libRadtran benchmark of [65]: test scores on the release’s split. Upper block, the release’s metrics over the three coefficients of every test row, and the RMSE of each coefficient; the first seven rows are the values published in Table 3 of the release for its model and its six baselines. Lower block, the retrieval scores at $\rho = 0.7$: the mean relative error of the top-of-atmosphere reflectance over the bands of a state, the 95th percentiles of $|\hat{\rho} - \rho|$ in points over all rows and on the physical domain above fT_{train} , and the share of failed inversions above $10^{-3}T_{\text{train}}$. Mean $\pm$ standard deviation over five seeds; 6S as the emulator is the release’s low-fidelity code scored as it stands.

radiance error and the tails: at $u = 1$ and above the radiance error falls by 0.02 percentage points or less on average and the tail above $10^{-3}T_{\text{train}}$ moves by less than 0.11 points on every split, while the all-band tail rises by 34 points.

8.2 The published libRadtran benchmark

On the benchmark of [65] (Section 7) the plain network stands near the release’s models in their own metrics (Table 2). Its RMSE over five seeds, 0.00669, lies between that of pKANrtm, 0.00619, and that of sRTMnet [10] as trained in the release, 0.00691, and below those of the release’s five other baselines, 0.00784 to 0.01394; a residual network tuned for this split in [85] reaches 0.00581.

The two weighted networks trade one of the release’s metrics against the others, on each of the five seeds. The floored weights at $u = 10^{-1}$ and the flattened weights with $\kappa = 1/2$ at $u = 10^{-2}$ lower the mean absolute error to 0.00083 and 0.00069, 0.45 and 0.37 of pKANrtm’s and below every model of the release, and the symmetric percentage error to 2.13% and 1.96%, against 3.91% for pKANrtm and 4.13% for the plain network. They raise the RMSE to 0.00785 and 0.00792, 1.27 and 1.28 times pKANrtm’s and about that of the release’s third model, and lower R^2 , which averages over the three coefficients, to 0.990, above only the release’s weakest baseline. The RMSE pools the three coefficients, and the spherical albedo sets it: the albedo’s error rises from 0.0094 to 0.0134 and 0.0136, while the errors of the path reflectance and of the transmittance fall by factors of 2.4 and 2.7 and of 3.1 and 3.9. The albedo’s retrieval weight vanishes with the transmission, as on the EMIT table (Section 8.1), and a single network with three outputs gives the capacity the other two weights ask for.

In the retrieval the weighted networks are better by a wide margin, again on every seed. The flattened network lowers the radiance error from 0.81% to 0.17%, the all-band 95th percentile from 3.21 to 0.48 points, the tails on the physical domain from 2.67, 2.38 and 2.31 to 0.44, 0.41 and 0.40 points, and the share of failed inversions above $10^{-3}T_{\text{train}}$ from 0.097% to 0.0002%; the floored

network at $u = 10^{-1}$ does nearly as well. 6S scored as the emulator has a radiance error of 20.6% and a tail of 75 points, so the networks learn a correction of it about two orders of magnitude smaller.

The result differs from that on the libRadtran table of Section 7, where no floor improved on the plain network in both the radiance error and the tails. The two settings differ in three ways at once: the benchmark trains on four and a half times as many states, its networks see the 6S coefficients and fit their residual, and they see one band of a state at a time, with the band among the inputs, where the networks of the table predict all bands of a state at once. The weighted networks were also still improving when training stopped: their lowest validation objective came at epochs 78 and 79 of 80 on every seed, against epochs 16 to 28 for the plain network.

Draft, results pending: a second set of runs on this benchmark, ten seeds at 300 epochs with five hidden layers of 512 units, adds the flattened weights with the albedo left unweighted, a mixture with the plain objective, a kernel correction of each network’s residuals fitted on the same weights, an out-of-distribution split that tests on the states above the 0.85 quantile of the aerosol optical depth or of the water-vapour column, as in [85] (the release reports an RMSE of 0.01022 for pKANrtm on its own split of this kind), and ablations in κ and u . Its results will replace the last paragraph.

8.3 Kernel regression fitted on the retrieval error

For kernel ridge regression the estimator of Section 5 is the truncated reweighting of [61]: for each component and band,

$$\min_f \sum_i w_i (f(x_i) - y_i)^2 + \lambda \|f\|^2, \quad \alpha_b = (K + \lambda W_b^{-1})^{-1} y_b,$$

with the entry weights of J_u (floored) or K_u (cut) at the true transmission. The weights change from band to band, so every band is its own solve, and to keep 285 solves per component affordable every arm is fitted on the same 6,000 training rows, the subsample on which the kernel’s parameters are chosen, with the isotropic Matérn kernel and the smoothness, length scale and nugget the forward criterion chooses on the validation block. The weights are scaled to mean one in each band, so that the nugget means the same in every arm, and the plain arm is the same solve with $w = 1$; the two forms of the normal equations, $(K + \lambda W_b^{-1})\alpha_b = y_b$ and $(W_b K + \lambda I)\alpha_b = W_b y_b$, give the same objectives to 10^{-13} on 300 rows. On the table without its two failed evaluations, over ten splits, the plain fit has a radiance error of 0.28% and tails of 24.8, 8.6 and 5.0 points on the physical domain (medians), against 0.075% and 1.39 points above $10^{-3}T_{\text{train}}$ for the input-scaled kernel of Section 8.5, which has its own length scales and is fitted on every training row.

The floored weights at $u = 10^{-1}$ lower every physical tail on every split, by 15, 13 and 9% at the median of the paired ratios, lower their own objective on held-out states on every split (median factor 0.75) and the share of failed inversions above $10^{-3}T_{\text{train}}$ from 0.67% to 0.42%, and raise the radiance error on every split, by 12%. At $u = 10^{-2}$ the radiance error rises by 27% and the tail above $10^{-12}T_{\text{train}}$ falls by 28%, while the one above $10^{-2}T_{\text{train}}$ rises by 4%, each on every split, and the objective falls to 0.28 of the plain fit’s. At $u = 10^{-3}$ the objective falls to 0.08 of the plain fit’s, and the radiance error rises by 44% and the tails above 10^{-3} and $10^{-2}T_{\text{train}}$ by 28 and 53%, again on every split. The cut weights at 10^{-2} lower the tails above 10^{-12} and $10^{-3}T_{\text{train}}$ by 61 and 23% and multiply the radiance error by 14. The kernel repeats what the networks of Section 8.1 show: its objective on held-out states falls at every floor, and the tails follow it only at the floor that keeps the most effective entries.

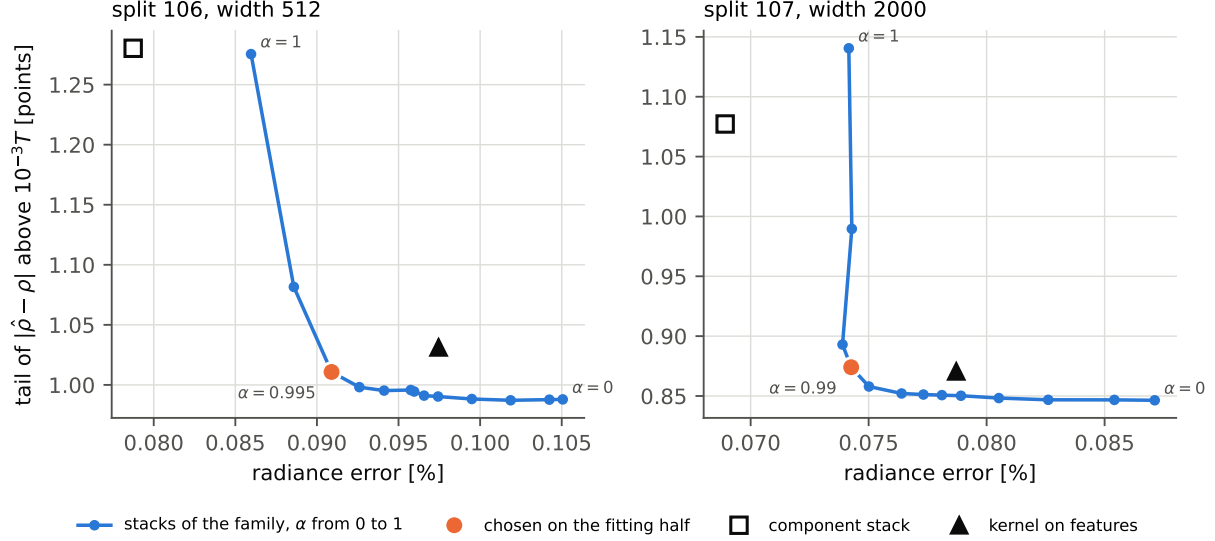


Figure 2: Stacks of the family of Proposition 12 on one held-out half of two EMIT runs, from the first-order retrieval error ($\alpha = 0$) to the forward error at $\rho = 0.7$ ($\alpha = 1$): the radiance error against the 95th percentile of the retrieval error on the physical domain above $10^{-3}T_{\text{train}}$. The orange point is the stack whose α is chosen on the other half; the square is the component stack and the triangle the kernel on learned features.

8.4 Stacks chosen on the retrieval error

The test states of each run are split in two halves; the weights of a stack are fitted on one half and scored on the other, both ways, so that no state enters both the fit and its evaluation and no member saw a test state. The *component stack* has one simplex per component fitted on the mean relative error of that component, the objective of the stack of [85]; the *retrieval stack* fits the four simplices jointly on the first-order retrieval error $\sum (v_\rho^\top e)^2$ at $\rho = 0.7$ over the entries of the physical domain with $t \geq 10^{-3}T_{\text{train}}$.

On the EMIT table, over twenty runs (ten splits at network widths 512 and 2000) and 40 held-out halves, the component stack reproduces the reversal: its radiance error is lower than that of the kernel on features on all 40 halves and its tail on the physical domain above $10^{-3}T_{\text{train}}$ larger on all 40. The retrieval stack has a tail below the component stack’s on 39 of the 40 halves and a higher radiance error on all 40. Along the family of Proposition 12 the tail stays nearly flat while the radiance error falls until the weight on the forward error is close to one (Figure 2).

With the mixture chosen on the fitting half alone, as the lowest radiance error among the objectives whose tail on that half is no larger than that of the best single member, the chosen weight on the forward error is at least 0.9 on 36 of the 40 halves (median 0.99), and the stack has a lower radiance error than the kernel on features on 33 of the 40 held-out halves and a lower tail than the component stack on 39, by 25% at the median (quartiles 20% and 29%, at most 39%). Its tail lies between 0.06 points below and 0.22 points above that of the kernel on features. Refitted on the floored criterion instead, on every entry with $0 \leq s < 1$ and with t floored at $10^{-3}T_{\text{train}}$, the stacks of splits 106–110 change their held-out tail by a factor between 0.995 and 1.014.

On the libRadtran table there is nothing to remove. Over the ten splits the component stack has a lower radiance error than the kernel on features on all 20 held-out halves and a smaller tail on 19, and the stack chosen by the same rule changes its tail by a factor between 0.93 and 1.05 and keeps

it below the feature kernel’s on all 20. Outside band B10 the density $g = t^2/q^4$ of Proposition 5 varies over a factor of 42 on this table, so the change of measure has little to reorder.

8.5 Kernel parameters chosen on the retrieval error

The kernel regressions of Section 7 choose their parameters on the validation block: the smoothness $\nu \in \{1.5, 2.5\}$, the length scale among four multiples of the median distance and the nugget among four values, and for the input-scaled kernel the relative length scales of the six inputs, by a coordinate search. The same grid and search can be run on a criterion for the retrieval instead of the forward error. The criteria for such choices are computed per component on the validation block, as the relative root-mean-square error weighted as in J_u : with the weights floored at uT_{train} (the *floored* criterion) or cut there, zero below (the *cut* criterion, with the weights v_τ of Proposition 8). The kernel is fitted with the chosen parameters on the plain squared error, and every run also chooses on the forward error, so that the differences are paired.

For the input-scaled kernel the retrieval criteria choose different parameters, and the choice pays in the tail. On the EMIT table without its two failed evaluations, the cut criterion at $u = 10^{-2}$ lowers the tail above $10^{-3}T_{\text{train}}$ from 1.39 to 0.72 points and the all-band tail from 33 to 5.8 points, on each of the ten splits, and nearly removes the failed inversions above $10^{-3}T_{\text{train}}$ (0.23% to 0.002%), while the radiance error rises from 0.075% to 0.34%. The floored criterion at 10^{-2} and the cut one at 10^{-3} do about as well. The chosen metric moves toward the water-vapour column: its share of the input weight rises from 0.27 to 0.70 for the direct flux, from 0.26 to 0.54 for the diffuse flux and from 0.14 to 0.42 for the path radiance (medians over the ten splits). On the table as provided, split 105, whose validation block holds one of the failed evaluations, is where every choice goes wrong, the cut ones included, since that state’s collapsing fluxes near 830 nm lie above the floor (Section 5.2); the floored criterion there picks a metric that raises the tail above $10^{-3}T_{\text{train}}$ by 18 points. For the isotropic kernel, which has no metric to move, every retrieval criterion raises the tail above $10^{-3}T_{\text{train}}$ on average, by 0.03 to 0.04 points at $u = 10^{-1}$ and by up to 6 points at 10^{-3} .

Choosing among a grid of kernels costs little in effective sample size, as Proposition 11 says. The input-scaled kernel chosen on the cut criterion at $u = 10^{-2}$ ends at about the radiance error of the networks of Section 8.1, 0.34% against 0.35% for the plain network and 0.34% for the network floored at $u = 10^{-1}$, lower than each on five splits of ten. Its tail above $10^{-3}T_{\text{train}}$ is 0.72 points against 3.5 and 2.2, lower on every split, about a fifth of the plain network’s and a third of the floored network’s (median ratios 0.20 and 0.33).

8.6 The stopping epoch

The stopping epoch of the plain network is a choice among the epochs of one training run, made here on the criteria of Section 8.5. On the EMIT table without its two failed evaluations, chosen by the cut criterion at $u = 10^{-1}$ instead of the plain validation loss, the stopping epoch lowers the radiance error on every split, by 0.005 percentage points on average, and the tail above $10^{-3}T_{\text{train}}$ on every split, by 0.06 points; the epoch has little to give. The criterion stops the two fluxes later, between epochs 141 and 150 against 106 to 150 for the plain rule, and at $u = 10^{-2}$ the cut and the floored criteria give the same small gains on every split. On the table as provided the choice at $u = 10^{-2}$ goes wrong on split 105, whose validation block holds one of the failed evaluations and whose entries both criteria count at that floor: the cut criterion stops the two fluxes at epochs 83 and 52 and the floored one at epochs 18 and 6, against 150 and 135 for the plain rule, and the radiance error rises by 0.26 and 1.24 percentage points.

9 Discussion

The retrieval error of an emulator is the forward error seen through a change of measure that favors small transmissions, and a table with entries deep in absorption bands makes the two rankings disagree. Fitting on the retrieval error follows the change of measure as far as the floor lets it, and the floor sets the price. On the EMIT table a floor at a tenth of the median transmission keeps enough effective entries for the fit to generalize, and the network trained on it improves on the plain one in every tail on every split; lower floors still lower their objective on held-out states but give up radiance error and, at the lowest, the tail. Flattening is the better way to temper the weights. The square root of the weights floored at a hundredth keeps the order of the small transmissions that a floor at a tenth erases and removes most of their range, and the network trained on it improves on the plain one in the radiance error and in every tail, on every split; the albedo, whose retrieval weight vanishes with the transmission below the floor, is the one component it fits less well. Kernel ridge regression on the floored weights moves as the networks do, so what is learned here concerns the weights and not the model class. The floor is therefore a parameter to be chosen on held-out states, and the lowest floor a fit can afford is set by the effective sample size of its weights rather than by the floor at which the retrieval is scored. A floored weight also magnifies whatever lies below the floor. Two failed evaluations of the code, two states in twenty-three thousand, turn the network at that floor from better than the plain one in every tail on every split to worse in the radiance error and the tail on every split, and a table meant for weighted training should first be screened for entries with zero transmission. Choosing on the retrieval error keeps the forward fit and moves only what the change of measure needs to move. For the stacks that removes most of the reversal, and choosing the mixture near the forward end of the family keeps the radiance error below that of the best single member on 33 of the 40 halves.

The first-order bound behind every criterion here is accurate where the component errors are small relative to the transmission, and the floors keep the criteria finite where they are not; the constrained retrieval and the transfer inequality of [85] cover the rest. The criterion is taken at $\rho = 0.7$ or over three reflectances; a retrieval over a different range of reflectances changes the metric but not the argument.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital; both entities supported this work and are named above. These entities and the public funders had no role in the study design, analysis, manuscript preparation, or decision to publish.

AI-assisted contributions

Claude Code (Anthropic) derived and checked the propositions, implemented and ran the experiments, produced the tables and figures from the run records, and drafted the manuscript. The author supplied the data, set the questions, and reviewed the results, the proofs and the text.

Data and code availability

The repository <https://github.com/yspennstate/rt-emulators-for-retrieval> holds the code, one JSON record per run with the digests of its inputs, the scripts that produce every table and figure from those records, and this manuscript. The EMIT table is the lookup table of [85], supplied by the Jet Propulsion Laboratory and available from the author on request, subject to the permission of the Jet Propulsion Laboratory. The libRadtran table and the benchmark of Section 8.2 are built from the public release accompanying [65].

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